# Prodigy Anesthesia Quick Review

## Electroconvulsive Therapy

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| What is electroconvulsive therapy (ECT)? | ECT is the induction of a generalized seizure via electrodes placed on the cranium.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 127.* |
| What is the most common complication of ECT? | Prolonged seizure, which is easily treated with benzodiazepines.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1268.* |
| What psychiatric disorders is ECT used to treat? | Schizoaffective disorders, bipolar disorders, and major depression disorders that are resistant to medical therapy.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1267.* |
| What are the two most common causes of death related to electroconvulsive therapy? | Cardiac dysrhythmias and myocardial infarction are the two most common causes of death related to ECT treatment.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 537-538.* |
| What are the hemodynamic effects of ECT? | ECT treatments result in stimulation of the autonomic nervous system resulting in a short-lived bradycardia followed by a surge in endogenous catecholamines causing potentially dangerous hypertension and tachycardia.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1277.* |
| Historically, what has been the preferred intravenous anesthetic agent for ECT? Why? | Methohexital has been used the most. All intravenous anesthetics reduce seizure activity, but methohexital reduces it the least.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1268.* |
| What type of anesthesia is administered for ECT? | General anesthesia is induced with an intravenous agent, succinylcholine is administered to provide muscle relaxation in preparation for the seizure, and positive-pressure ventilation by mask is performed via mask (or via ETT if the patient is at risk for aspiration).<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1268.* |
| What are the anesthetic goals for ECT? | Provide airway support, prevent any recall of the procedure, and attenuate the hemodynamic response to ECT.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1268.* |

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